PROBLEM SOLVING THROUGH AI&ML

Course Code: CSA 601 **Credit Units: 03**
Total Hours: 45

Course Objective:
The main objective is to teach utility of basic methods of Artificial Intelligence and Machine learning in various case studies in industries, including:, planning, search, localization of errors, tracking, prediction and control of the system.

Course Contents:

Module I: Introduction: (9 Hours)
Machine Learning Applications across Industries - Healthcare, Retail, Financial Services, Manufacturing, Hospitality

Module II: Churn Analysis and Fraud Detection: (9 Hours)
Churn Analysis and Prediction (Survival Modelling): Churn Prediction, Credit card Fraud Analysis: Imbalanced Data, Neural Network

Module III: Handling Text Data: (9 Hours)
Handling Text Data, Bag-of-words, Regular Expressions, Sentence Splitting and Tokenization, Punctuations and Stop words, Incorrect spellings, Properties of words and Word cloud, Sentiment Analysis (Case Study)

Module IV: Forecasting and Prediction: (9 Hours)
Forecasting models, Trend analysis, Cyclical and Seasonal analysis, Smoothing; Moving averages; Box-Jenkins, Holt-winters, Auto-correlation; ARIMA, Examples: Applications of Time Series in financial markets

Module V: Recommender Systems: (9 Hours)
Recommendation Systems and Collaborative filtering, User based, Item Based, Singular value decomposition–based recommenders, Customer Segmentation and Value, Social Networking Analysis

Course Outcomes:
At the end of this course, students will learn:
- To study concepts of various real-time case studies with respect to machine learning.
- To understand the utility of ML techniques in industrial use cases.

Examination Scheme:

Components	A	CT	S/V/Q/HA	EE
Weightage (%)	5	15	10	70

CT: Class Test, HA: Home Assignment, S/V/Q: Seminar/Viva/Quiz, EE: End Semester Examination; A: Attendance

Text & References:
Text:
- Machine Learning by Saikat Dutt , Subramanian Chandramouli, Pearson Education; First edition (1 October 2018)
- Chandra S.S.V, Artificial Intelligence and Machine Learning, Prentice Hall India Learning Private Limited; 1 edition (2014)

References:

- Tom M. Mitchell, ―Machine Learning, McGraw-Hill Education (India) Private Limited, 2013.
- Ethem Alpaydin, ―Introduction to Machine Learning (Adaptive Computation and Machine Learning), The MIT Press 2004.
- Stephen Marsland, ―Machine Learning: An Algorithmic Perspective, CRC Press, 2009.
- Christopher M. Bishop, Pattern Recognition and Machine Learning.